

Put Option Basics

Much of the same terminology that is applied to call options also pertains to put options. *Underlying security*, *striking price*, and *expiration date* are all terms that have the same meaning for puts as they do for calls. The expiration dates of listed put options agree with the expiration dates of the calls on the same underlying stock. In addition, puts and calls have the same striking prices. This means that if there are options at a certain strike, say on a particular underlying stock that has both listed puts and calls, both calls at 50 and puts at 50 will be trading, regardless of the price of the underlying stock. Note that it is no longer sufficient to describe an option as an “XYZ July 50.” It must also be stated whether the option is a put or a call, for an XYZ July 50 call and an XYZ July 50 put are two different securities.

In many respects, the put option and its associated strategies will be very nearly the opposite of corresponding call-oriented strategies. However, *it is not correct to say that the put is exactly the opposite of a call*. In this introductory section on puts, the characteristics of puts are described in an attempt to show how they are similar to calls and how they are not.

PUT STRATEGIES

In the simplest terms, *the outright buyer of a put is hoping for a stock price decline* in order for his put to become more valuable. If the stock were to decline well below the striking price of the put option, the put holder could make a profit. The holder of the put could buy stock in the open market and then exercise his put to sell that stock for a profit at the striking price, which is higher.

Example: If XYZ stock is at 40, an XYZ July 50 put would be worth at least 10 points, for the put grants the holder the right to sell XYZ at 50—10 points above its current price.

On the other hand, if the stock price were *above* the striking price of the put option at expiration, the put would be worthless. No one would logically want to exercise a put option to sell stock at the striking price when he could merely go to the open market and sell the stock for a higher price. Thus, *as the price of the underlying stock declines, the put becomes more valuable*. This is, of course, the opposite of a call option's price action.

The meaning of in-the-money and out-of-the-money are altered when one is speaking of put options. *A put is considered to be in-the-money when the underlying stock is below the striking price of the put option; it is out-of-the-money when the stock is above the striking price*. This, again, is the opposite of the call option. If XYZ is at 45, the XYZ July 50 put is in-the-money and the XYZ July 50 call is out-of-the-money. However, if XYZ were at 55, the July 50 put would be out-of-the-money while the July 50 call would be in-the-money. The broad definition of an in-the-money option as "an option that has intrinsic value" would cover the situation for both puts and calls. Note that a put option has intrinsic value when the underlying stock is below the striking price of the put. That is, the put has some "real" value when the stock is below the striking price.

The intrinsic value of an in-the-money put is merely the difference between the striking price and the stock price. Since the put is an option (to sell), it will generally sell for more than its intrinsic value when there is time remaining until the expiration date. This excess value over its intrinsic value is referred to as the *time value premium*, just as is the case with calls.

Example: XYZ is at 47 and the XYZ July 50 put is selling for 5, the intrinsic value is 3 points (50 – 47), so the time value premium must be 2 points. The time value premium of an in-the-money put option can always be quickly computed by the following formula:

$$\begin{array}{l} \text{Time value premium} \\ \text{(in-the-money put)} \end{array} = \text{Put option} + \text{Stock price} - \text{Striking price}$$

This is not the same formula that was applied to in-the-money call options, although it is always true that the time value premium of an option is the excess value over intrinsic value.

$$\begin{array}{l} \text{Time value premium} \\ \text{(in-the-money call)} \end{array} = \text{Call option} + \text{Striking price} - \text{Stock price}$$

If the put is out-of-the-money, the entire premium of the put is composed of time value premium, for the intrinsic value of an out-of-the-money option is always zero. The *time value premium of a put is largest when the stock is at the striking price of the put*. As the option becomes deeply in-the-money or deeply out-of-the-money, the time value

premium will shrink substantially. These statements on the magnitude of the time value premium are true for both puts and calls. Table 15-1 will help to illustrate the relationship of stock price and option price for both puts and calls. The reader may want to refer to Table 1-1, which described the time value premium relationship for calls. Table 15-1 describes the prices of an XYZ July 50 call option and an XYZ July 50 put option.

Table 15-1 demonstrates several basic facts. As the stock drops, the actual price of a call option decreases while the value of the put option increases. Conversely, as the stock rises, the call option increases in value and the put option decreases in value. Both the put and the call have their maximum time value premium when the stock is exactly at the striking price. However, *the call will generally sell for more than the put when the stock is at the strike*. Notice in Table 15-1 that, with XYZ at 50, the call is worth 5 points while the put is worth only 4 points. This is true in general, except in the case of a stock that pays a large dividend. This phenomenon has to do with the cost of carrying stock. More will be said about this effect later. Table 15-1 also describes an effect of put options that normally holds true: *An in-the-money put (stock is below strike) loses time value premium more quickly than an in-the-money call does*. Notice that with XYZ at 43 in Table 15-1, the put is 7 points in-the-money and has lost all its time value premium. But when the call is 7 points in-the-money, XYZ at 57, the call still has 2 points of time value premium. Again, this is a phenomenon that could be affected by the dividend payout of the underlying stock, but is true in general.

PRICING PUT OPTIONS

The same factors that determine the price of the call option also determine the price of the put option: price of the underlying stock, striking price of the option, time remaining until expiration, volatility of the underlying stock, dividend rate of the underlying stock, and the current risk-free interest rate (Treasury bill rate, for example). Market dynamics—supply, demand, and investor psychology—play a part as well.

Without going into as much detail as was shown in Chapter 1, the pricing curve of the put option can be developed. Certain facts remain true for the put option as they did for the call option. The rate of decay of the put option is not linear; that is, the time value premium will decay more rapidly in the weeks immediately preceding expiration. The more volatile the underlying stock, the higher will be the price of its options, both puts and calls. Moreover, the marketplace may at any time value options at a higher or lower volatility than the underlying stock actually exhibits. This is called implied volatility, as distinguished from actual volatility. Also, the put option is usually worth at least its intrinsic value at any time, and should be worth exactly its intrinsic value on the day that it expires. Figure 15-1 shows where one might expect the XYZ July 50 put to sell, for any

TABLE 15-1.
Call and put options compared.

XYZ Stock Price	XYZ July 50 Call Price	Call Intrinsic Value	Call Time Value Premium	XYZ July 50 Put Price	Put Intrinsic Value	Put Time Value Premium
40	.50	0	.50	9.75	10	-.25*
43	1	0	1	7	7	0
45	2	0	2	6	5	1
47	3	0	3	5	3	2
50	5	0	5	4	0	4
53	7	3	4	3	0	3
55	8	5	3	2	0	2
57	9	7	2	1	0	1
60	10.50	10	.50	.50	0	.50
70	19.75	20	-.25*	.25	0	.25

*A deeply in-the-money option may actually trade at a discount from intrinsic value in advance of expiration.

stock price, if there are 6 months remaining until expiration. Compare this with the similar pricing curve for the call option (Figure 15-2). Note that the *intrinsic value line* for the put option faces in the opposite direction from the intrinsic value line for call options; that is, it gains value as the stock falls below the striking price. This put option pricing curve demonstrates the effect mentioned earlier, that a put option loses time value premium more quickly when it is in-the-money, and also shows that an out-of-the-money put holds a great deal of time value premium.

THE EFFECT OF DIVIDENDS ON PUT OPTION PREMIUMS

The dividend of the underlying stock is a negative factor on the price of its call options. The opposite is true for puts. *The larger the dividend, the more valuable the puts will be.* This is true because, as the stock goes ex-dividend, it will be reduced in price by the amount of the dividend. That is, the stock will decrease in price and therefore the put will become more valuable. Consequently, the buyer of the put will be willing to pay a higher price for the put and the seller of the put will also demand a higher price. As with listed calls, listed puts are not adjusted for the payment of cash dividends on the underlying stock. However, the price of the option itself will reflect the dividend payments on the stock.

FIGURE 15-1.
Put option price curve.

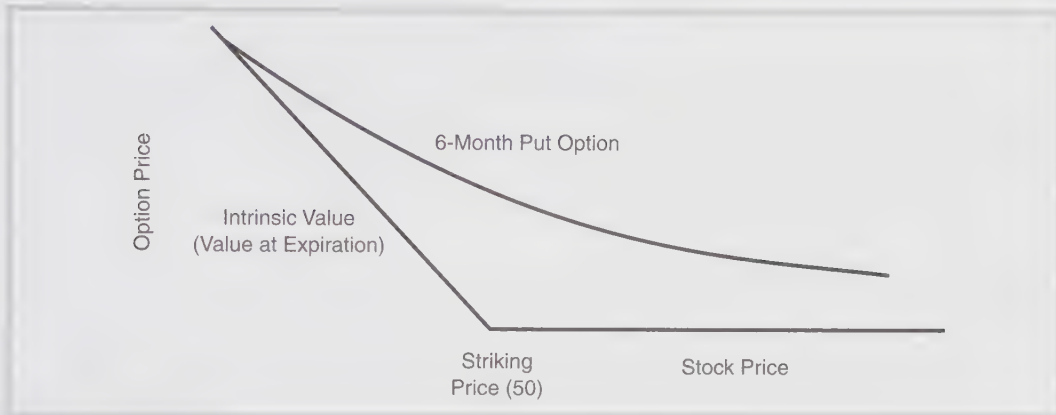
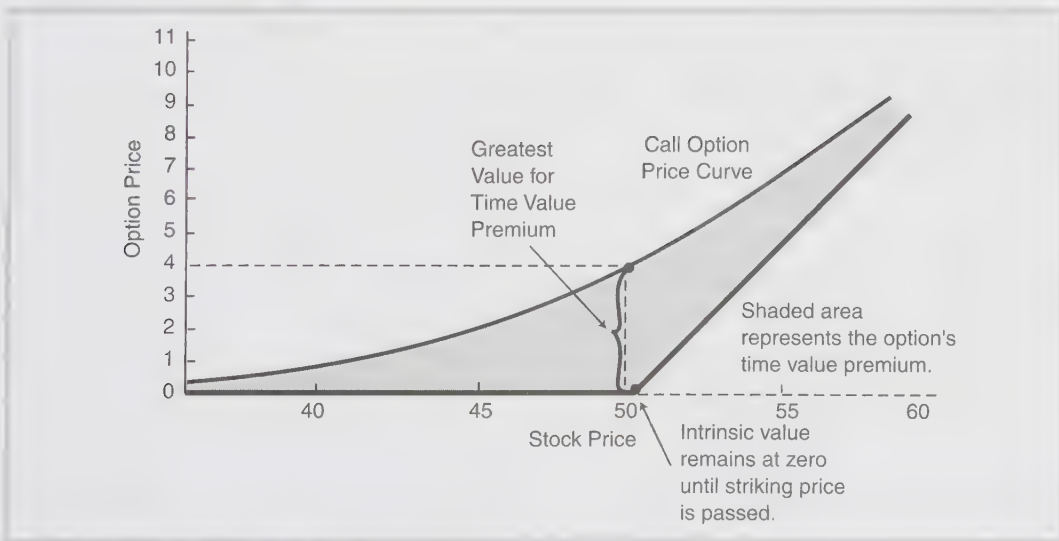


FIGURE 15-2.
Call option price curve.



Example: XYZ is selling for \$25 per share and will pay \$1 in dividends over the next 6 months. Then a 6-month put option with strike 25 should automatically be worth at least \$1, regardless of any other factor concerning the underlying stock. During the next 6 months, the stock will be reduced in price by the amount of its dividends—\$1—and if everything else remained the same, the stock would then be at 24. With the stock at 24, the put would be 1 point in-the-money and would thus be worth at least its intrinsic value

of 1 point. Thus, in advance, this large dividend payout of the underlying stock will help to increase the price of the put options on this stock.

On the day before a stock goes ex-dividend, the time value premium of an in-the-money put should be at least as large as the impending cash dividend payment. That is, if XYZ is 40 and is about to pay a \$.50 dividend, an XYZ January 50 put should sell for at least 10.50. This is true because the stock will be reduced in price by the amount of its dividend on the day of the ex-dividend.

EXERCISE AND ASSIGNMENT

When the holder of a put option exercises his option, he sells stock at the striking price. He may exercise this right at any time during the life of the put option. When this happens, *the writer of a put option with the same terms is assigned an obligation to buy stock at the striking price.* It is important to notice the difference between puts and calls in this case. The call holder exercises to buy stock and the call writer is obligated to sell stock. The reverse is true for the put holder and writer.

The methods of assignment via the OCC and the brokerage firm are the same for puts and calls; any fair method of random or first-in/first-out assignment is allowed. Stock commissions are charged on both the purchase and sale of the stock via the assignment and exercise.

When the holder of a put option exercises his right to sell stock, he may be selling stock that he currently holds in his portfolio. Second, he may simultaneously go into the open market and buy stock for sale via the put exercise. Finally, he may want to sell the stock in his short stock account; that is, he may short the underlying stock by exercising his put option. He would have to be able to borrow stock and supply the margin collateral for a short sale of stock if he chose this third course of action.

The writer of the put option also has several choices in how he wants to handle the stock purchase that he is required to make. *The put writer who is assigned must receive stock.* (The call writer who is assigned delivers stock.) The put writer may currently be short the underlying stock, in which case he will merely use the receipt of stock from the assignment to cover his short sale. He may also decide to immediately sell stock in the open market to offset the purchase that he is forced to make via the put assignment. Finally, he may decide to retain the stock that is delivered to him; he merely keeps the stock in his portfolio. He would, of course, have to pay for (or margin) the stock if he decides to keep it.

The mechanics as to how the put holder wants to deliver the stock and how the put writer wants to receive the stock are relatively simple. Each one merely notifies his brokerage firm of the way in which he wants to operate and, provided that he can meet the margin requirements, the exercise or assignment will be made in the desired manner.

ANTICIPATING ASSIGNMENT

The writer of a put option can anticipate assignment in the same way that the writer of a call can. *When the time value premium of an in-the-money put option disappears, there is a risk of assignment, regardless of the time remaining until expiration.* In Chapter 1, a form of arbitrage was described in which market-makers or firm traders, who pay little or no commissions, can take advantage of an in-the-money call selling at a discount to parity. Similarly, there is a method for these traders to take advantage of an in-the-money put selling at a discount to parity.

Example: XYZ is at 40 and an XYZ July 50 put is selling for 9.80—a .20 discount from parity. That is, the option is selling for .20 points below its intrinsic value. The arbitrageur could take advantage of this situation through the following actions:

1. Buy the July put at 9.80.
2. Buy XYZ common stock at 40.
3. Exercise the put to sell XYZ at 50.

The arbitrageur makes 10 points on the stock portion of the transaction, buying the common at 40 and selling it at 50 via exercise of his put. He paid 9.80 for the put option and he loses this entire amount upon exercise. However, his overall profit is thus .25 point, the amount of the original discount from parity. Since his commission costs are minimal, he can actually make a net profit on this transaction.

As was the case with deeply in-the-money calls, this type of arbitrage with deeply in-the-money puts provides a secondary market that might not otherwise exist. It allows the public holder of an in-the-money put to sell his option at a price near its intrinsic value. Without these arbitrageurs, there might not be a reasonable secondary market in which public put holders could liquidate.

Dividend payment dates may also have an effect on the frequency of assignment. For call options, the writer might expect to receive an assignment on the day the stock goes ex-dividend. The holder of the call is able to collect the dividend by so exercising. Things are slightly different for the writer of puts. He might expect to receive an assignment on the day after the ex-dividend date of the underlying stock. Since the writer of the put is obligated to buy stock, it is unlikely that anyone would put the stock to him until after the dividend has been paid. In any case, the writer of the put can use a relatively simple gauge to anticipate assignment near the ex-dividend date. If the time value premium of an in-the-money put is less than the amount of the dividend to be paid, the writer may often anticipate that he will be assigned immediately after the ex-dividend of the stock. An example will show why this is true.

Example: XYZ is at 45 and it will pay a \$.50 dividend. Furthermore, the XYZ July 50 put is selling at 5.25. Note that the time value premium of the July 50 put is 25 cents—less than the amount of the dividend, which is 50 cents. An arbitrageur could take the following actions:

1. Buy XYZ at 45.
2. Buy the July 50 put at 5.25.
3. Collect the 50-cent dividend (he must hold the stock until the ex-date to collect the dividend).
4. Exercise his put to sell XYZ at 50 (writer would receive assignment on the day after the ex-date).

The arbitrageur makes 5 points on the stock trades, buying XYZ at 45 and selling it at 50 via exercise of the put. He also collects the 50-cent dividend, making his total intake equal to 5.50 points. He loses the 5.25 points that he paid for the put but still has a net profit of 25 cents. Thus, *as the ex-dividend date of a stock approaches, the time value premium of all in-the-money puts on that stock will tend to equal or exceed the amount of the dividend payment.*

This is quite different from the call option. It was shown in Chapter 1 that the call writer only needs to observe whether the call was trading at or below parity, regardless of the amount of the dividend, as the ex-dividend date approaches. The put writer must determine if the time value premium of the put exceeds the amount of the dividend to be paid. If it does, there is a much smaller chance of assignment because of the dividend. In any case, the put writer can anticipate the assignment if he carefully monitors his position.

POSITION LIMITS

Recall that the position limit rule states that one cannot have a position of more than the limit of options on the same side of the market in the same underlying security. The limit varies depending on the trading activity and volatility of the underlying stock and is set by the exchange on which the options are traded. The actual limits are 13,500, 22,500, 31,500, 60,000, or 75,000 contracts, depending on these factors. One cannot have more than 75,000 option contracts on the bullish side of the market—long calls and/or short puts—nor can he have more than 75,000 contracts on the bearish side of the market—short calls and/or long puts. He may, however, have 75,000 contracts on each side of the market; he could simultaneously be long 75,000 calls and long 75,000 puts.

For the following examples, assume that one is concerned with an underlying stock whose position limit is 75,000 contracts.

Long 75,000 calls, long 75,000 puts—no violation; 75,000 contracts bullish (long calls) and 75,000 contracts bearish (long puts).

Long 38,000 calls, short 37,000 puts—no violation; total of 75,000 contracts bullish. Long 38,000 calls, short 38,000 puts—violation; total of 76,000 contracts bullish.

Money managers should be aware that these position limits apply to all “related” accounts, so that someone managing several accounts must total all the accounts’ positions when considering the position limit rule.

CONVERSION

Many of the relationships between call prices and put prices relate to a process known as a *conversion*. This term dates back to the over-the-counter option days when a dealer who owned a put (or could buy one) was able to satisfy the needs of a potential call buyer by “converting” the put to a call. This terminology is somewhat confusing, and the actual position that the dealer would take is little more than an arbitrage position. In the listed market, arbitrageurs and firm traders can set up the same position that the converter did.

The actual details of the conversion process, which must include the carrying cost of owning stock and the inclusion of all dividends to be paid by the stock during the time the position is held, are described later. However, it is important for the put option trader to understand what the arbitrageur is attempting to do in order for him to fully understand the relationship between put and call prices in the listed option market.

A conversion position has no risk. The arbitrageur will do three things:

1. Buy 100 shares of the underlying stock.
2. Buy 1 put option at a certain striking price.
3. Sell 1 call option at the same striking price.

The arbitrageur has no risk in this position. If the underlying stock drops, he can always exercise his long put to sell the stock at a higher price. If the underlying stock rises, his long stock offsets the loss on his short call. Of course, the prices that the arbitrageur pays for the individual securities determine whether or not a conversion will be profitable. At times, a public customer may look at prices in the newspaper and see that he could establish a position similar to the foregoing one for a profit, even after commissions. However, unless prices are out of line, the public customer would not normally be able to make a better return than he could by putting his money into a bank or a Treasury bill, because of the commission costs he would pay.

Without needing to understand, at this time, exactly what prices would make an attractive conversion, it is possible to see that it would not always be possible for the arbitrageur to do a conversion. The mere action of many arbitrageurs doing the same conversion would force the prices into line. The stock price would rise because arbitrageurs are buying the stock, as would the put price; and the call price would drop because of the preponderance of sellers.

When this happens, another arbitrage, known as a *reversal* (or *reverse conversion*), is possible. In this case, the arbitrageur does the opposite: He shorts the underlying stock, sells 1 put, and buys 1 call. Again, this is a position with no risk. If the stock rises, he can always exercise his call to buy stock at a lower price and cover his short sale. If the stock falls, his short stock will offset any losses on his short put.

The point of introducing this information, which is relatively complicated, at this place in the text is to demonstrate that *there is a relationship between put and call prices, when both have the same striking price and expiration date*. They are not independent of one another. If the put becomes “cheap” with respect to the call, arbitrageurs will move in to do conversions and force the prices back in line. On the other hand, if the put becomes expensive with relationship to the call, arbitrageurs will do reversals until the prices move back into line.

Because of the way in which the carrying cost of the stock and the dividend rate of the stock are involved in doing these conversions or reversals, two facts come to light regarding the relationship of put prices and call prices. Both of these facts have to do with the carrying costs incurred during the conversion. First, *a put option will generally sell for less than a call option when the underlying stock is exactly at the striking price*, unless the stock pays a large dividend. In the older over-the-counter option market, it was often stated that the reason for this relationship was that the demand for calls was larger than the demand for puts. This may have been partially true, but certainly it is no longer true in the listed option targets, where a large supply of both listed puts and calls is available through the OCC. Arbitrageurs again serve a useful function in increasing supply and demand where it might not otherwise exist. The second fact concerning the relationship of puts and calls is that *a put option will lose its time value premium much more quickly in-the-money than a call option will* (and, conversely, a put option will generally hold out-of-the-money time value premium better than a call option will). Again, the conversion and reversal processes play a large role in this price action phenomenon of puts and calls. Both of these facts have to do with the carrying costs involved in the conversion.

In the chapter on Arbitrage, exact details of conversions and reversals will be spelled out, with specific reasons why these procedures affect the relationship of put and call prices as stated above. However, at this time, it is sufficient for the reader to understand that there is an arbitrage process that is quite widely practiced that will, in fact, make true the foregoing relationships between puts and calls.